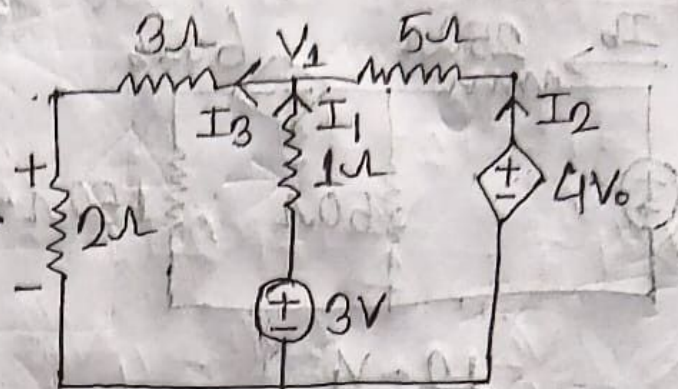
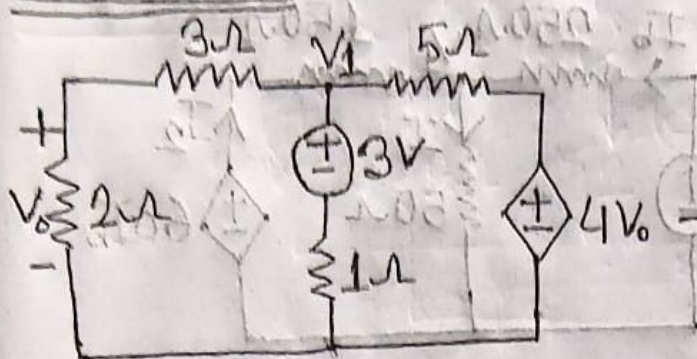


NODAL ANALYSIS

Prob 3.8



$$V_0 = 2 \times I_3 = 2I_3$$

$$4V_0 = 4 \times 2I_3 = 8I_3$$

$$= 8 \left(\frac{V_1 - 0}{5} \right)$$

Apply KCL at Node 1,

$$I_1 + I_2 = I_3$$

$$\frac{3 - V_1}{1} + \frac{4V_0 - V_1}{5} = \frac{V_1 - 0}{5}$$

$$\Rightarrow \frac{3 - V_1}{1} + \frac{8V_1 - 5V_1}{5} - \frac{V_1}{5} = 0$$

$$\Rightarrow \frac{3 - V_1}{1} + \frac{3V_1}{5} - \frac{V_1}{5} = 0$$

$$\Rightarrow \frac{75 - 25V_1 + 3V_1 - 5V_1}{25} = 0$$

$$\Rightarrow -27V_1 + 75 = 0$$

$$\Rightarrow 27V_1 = 75$$

$$\therefore V_1 = 2.78V$$

Find $V_0 = ?$

$$\therefore V_0 = 2I_3$$

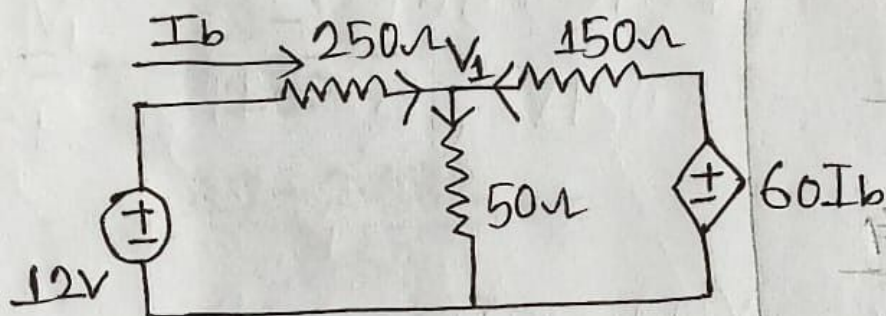
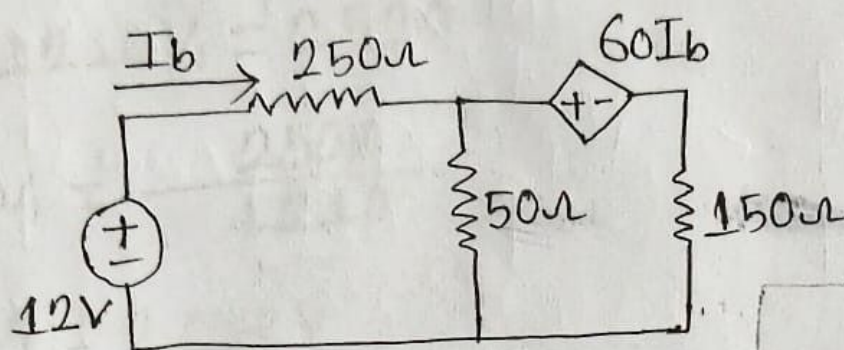
$$= 2 \times \frac{V_1 - 0}{5}$$

$$= 2 \times \left(\frac{2.78}{5} \right)$$

$$= 1.112V$$

Ans

Prob. 3.9



$$I_b = \frac{12 - V_1}{250}$$

$$60I_b = 60 \times \frac{12 - V_1}{250}$$

$$= \frac{720 - 60V_1}{250}$$

Apply KCL at node V_1 ,

$$\frac{12 - V_1}{250} + \frac{\left(\frac{720 - 60V_1}{250}\right) - V_1}{150} - \frac{V_1}{50} = 0$$

$$\Rightarrow \frac{12 - V_1}{250} + \frac{720 - 60V_1 - 250V_1}{37500} - \frac{V_1}{50} = 0$$

$$\Rightarrow -1210V_1 + 2520 = 0$$

$$\Rightarrow 1210V_1 = 2520$$

$$\Rightarrow V_1 = \frac{2520}{1210}$$

$$\therefore V_1 = 2.082V$$

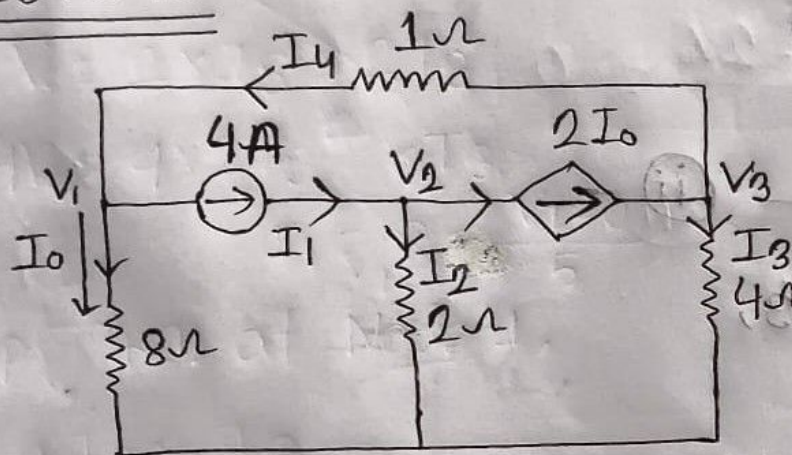
$$I_b = \frac{12 - V_1}{250}$$

$$= \frac{12 - 2.082}{250}$$

$$\therefore I_b = 0.039A$$

Ans

Prob: 3.10



$$I_0 = \frac{V_1}{8}$$

Find, $I_0 = ?$

$$\therefore 2I_0 = \frac{2V_1}{8}$$

Apply KCL at node V_1 ,

$$I_4 = I_0 + I_1$$

$$\Rightarrow \frac{V_3 - V_1}{1} = \frac{V_1}{8} + 4 \quad \text{(iii)}$$

$$\Rightarrow \frac{V_1 + 32 - 8V_3 + 8V_1}{8} = 0$$

$$\Rightarrow 9V_1 - 8V_3 = -32 \quad \text{(i)}$$

(i)

V80.2 - IV

80,2-01

250

A1040

of 2: don't

11

$$V_1 = -32V$$

$$V_2 = 24V$$

$$V_3 = -32V'$$

$$\therefore I_0 = \frac{V_1}{8} = \frac{-32}{8}$$

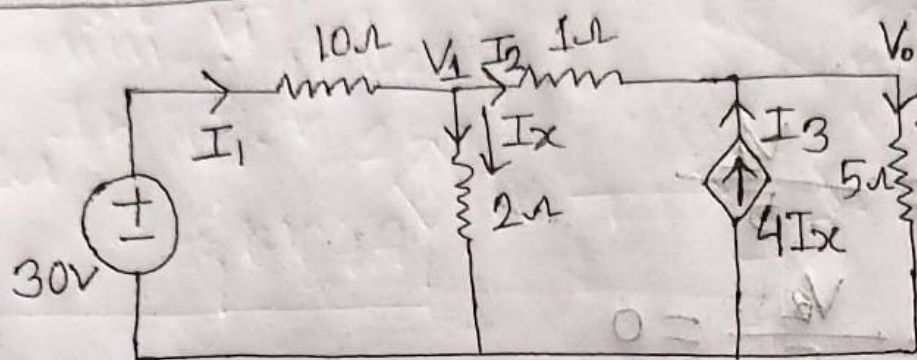
2-4A Am

$$V_1 = -32 \text{ V}$$

$$V_2 = 24V$$

$$V_3 = -32V'$$

Prob: 3.12



Find, $V_0 = ?$

$$I_x = \frac{V_1}{2}$$

$$\therefore 4I_x = \frac{4V_1}{2}$$

Apply KCL at Node V_1 ,

$$I_1 = I_x + I_2$$

$$\Rightarrow \frac{30 - V_1}{10} = \frac{V_1}{2} + \frac{V_1 - V_0}{1}$$

$$\Rightarrow \frac{30 - V_1}{10} - \frac{V_1}{2} - V_1 + V_0 = 0$$

$$\Rightarrow \frac{30 - V_1 - 5V_1 - 10V_1 + 10V_0}{10} = 0$$

$$\Rightarrow -16V_1 + 10V_0 = -30 \quad \text{--- (i)}$$

~~Apply~~

Apply KCL at node V_0 ,

$$I_2 + I_3 = I_4$$

$$\Rightarrow \frac{V_1 - V_0}{1} + 4I_x = \frac{V_0}{5}$$

$$\Rightarrow V_1 - V_0 + \frac{4V_1}{2} - \frac{V_0}{5} = 0$$

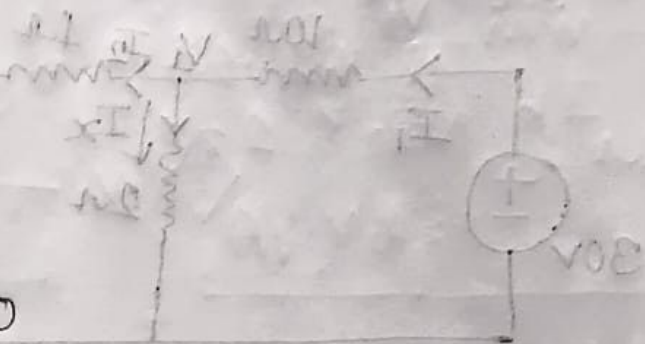
$$\Rightarrow \frac{10V_1 - 10V_0 + 20V_1 - 2V_0}{10} = 0$$

$$\Rightarrow 30V_1 - 12V_0 = 0 \quad \text{--- (ii)}$$

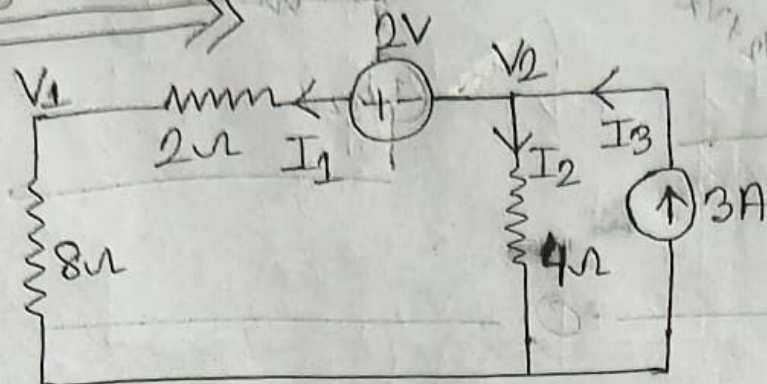
$$\therefore V_1 = -3.33V$$

$$\therefore V_0 = -8.33V$$

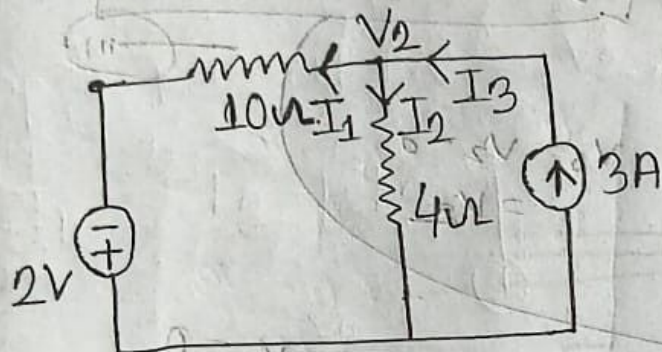
Ans



Prob 3.13



$$V_1 = 8I_1$$



Apply KCL at node V_2 ,

$$I_3 = I_1 + I_2$$

$$\Rightarrow 3 = \frac{V_2 + 2}{10} + \frac{V_2}{4}$$

$$\Rightarrow 3 - \frac{V_2 + 2}{10} - \frac{V_2}{4} = 0$$

$$\Rightarrow \frac{60 - 2V_2 - 4 - 5V_2}{20} = 0$$

$$\Rightarrow 56 - 7V_2 = 0$$

$$\Rightarrow V_2 = \frac{56}{7} = 8V$$

$$\therefore V_2 = 8V$$

$$\therefore V_1 = 8I_1$$

$$= 8 \times \frac{V_2 + 2}{10}$$

$$= 8 \times \frac{8 + 2}{10}$$

$$\therefore V_1 = 8V$$

$$I_1 = \frac{V_2 + 2}{10} = \frac{8 + 2}{10} = 1A$$

$$I_2 = \frac{V_2 - 0}{4} = \frac{8}{4} = 2A$$

$$I_3 = 3A$$

$$I_1 = \frac{V_2 + 2}{10} = \frac{8 + 2}{10} = 1A$$

$$I_2 = \frac{V_2 - 0}{4} = \frac{8}{4} = 2A$$

$$I_3 = 3A$$

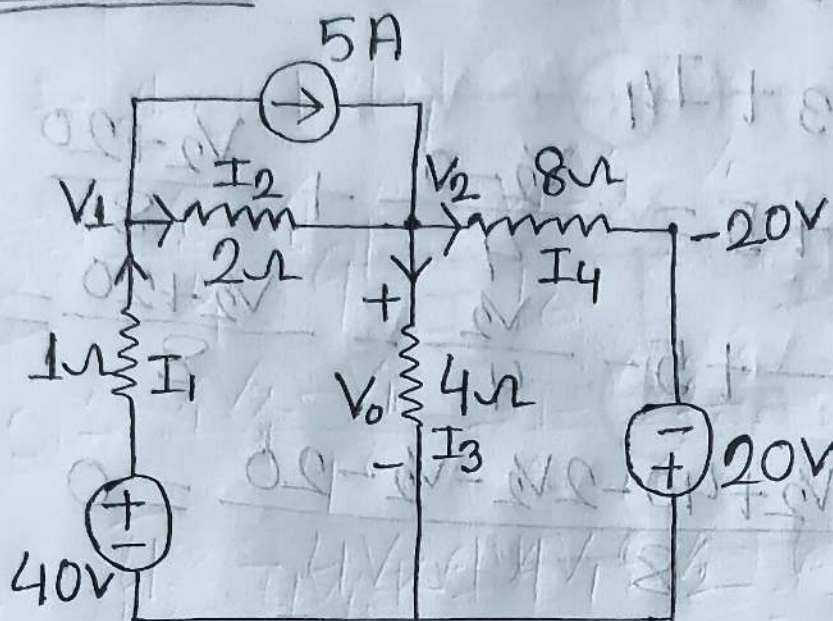
$$I_1 = \frac{V_2 + 2}{10} = \frac{8 + 2}{10} = 1A$$

$$I_2 = \frac{V_2 - 0}{4} = \frac{8}{4} = 2A$$

$$I_3 = 3A$$

$$I_1 = \frac{V_2 + 2}{10} = \frac{8 + 2}{10} = 1A$$

Prob 3.14



$$V_0 = V_2 - 0$$

$$\Rightarrow V_0 = V_2$$

Apply KCL at node V_1 ,

$$I_1 = I_2 + 5$$

$$\frac{40 - V_1}{1} = \frac{V_1 - V_2}{2} + 5$$

$$\Rightarrow \frac{40 - V_1}{1} - \frac{V_1 - V_2}{2} - 5 = 0$$

$$\Rightarrow \frac{80 - 2V_1 - V_1 + V_2 - 10}{2} = 0$$

$$\Rightarrow -3V_1 + V_2 = -70 \quad \text{--- (i)}$$

Apply KCL at node V_2 , 11.8 down

$$I_2 + 5 = I_3 + I_4$$

$$\Rightarrow \frac{V_1 - V_2}{2} + 5 = \frac{V_2 - 0}{4} + \frac{V_2 + 20}{8}$$

$$\Rightarrow \frac{V_1 - V_2}{2} + 5 - \frac{V_2}{4} - \frac{V_2 + 20}{8} = 0$$

$$\Rightarrow \frac{4V_1 - 4V_2 + 40 - 2V_2 - V_2 - 20}{8} = 0$$

$$\Rightarrow 4V_1 - 7V_2 = -20 \quad \text{--- (ii)}$$

$$\therefore V_1 = 30V$$

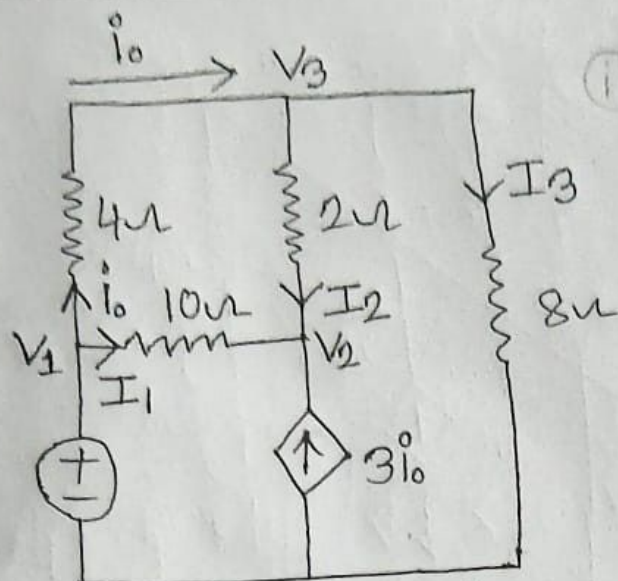
$$\therefore V_2 = 20V$$

$$V_o = V_2$$

$$\therefore V_o = 20V$$

Ans

Prob: 3.17



Find current i_0 ,

Apply KCL at Node V_2 ,

$$I_1 + I_2 + 3i_0 = 0$$

$$\Rightarrow \frac{60 - V_2}{10} + \frac{V_3 - V_2}{2} + \frac{180 - 3V_3}{4} = 0 \text{ mA}$$

$$\Rightarrow \frac{120 - 2V_2 + 10V_3 - 10V_2 + 900 - 15V_3}{20} = 0$$

$$\Rightarrow -12V_2 - 5V_3 = -1020 \quad \text{--- (1)}$$

Apply KCL at Node V_3 ,

$$i_0 = I_2 + I_3$$

$$\Rightarrow \frac{60 - V_3}{4} = \frac{V_3 - V_2}{2} + \frac{V_3 - 0}{8}$$

$$\Rightarrow \frac{60 - V_3}{4} - \frac{V_3 - V_2}{2} - \frac{V_3 - 0}{8} = 0$$

$$0 = \frac{8V - 8V_1 + 8V_1 - 8V_2 - 0.01}{8} \quad \leftarrow$$

$$(11) \quad 0.01 = 8V_1 - 8V_2 \quad \leftarrow$$

$$V_1 = 0.88V$$

$$V_2 = 0.88V$$

$$\frac{8V - 0.01}{4} = i_0$$

$$\frac{8V - 0.01}{4} = i_0$$

$$i_0 = \frac{V_1 - V_3}{4}$$

$$3i_0 = 3 \times \frac{60 - V_3}{4}$$

$$= \frac{180 - 3V_3}{4}$$

$$\Rightarrow \frac{120 - 2V_3 - 4V_3 + 4V_2 - V_3}{8} = 0$$

$$\Rightarrow 4V_2 - 7V_3 = -120 \quad \text{--- (ii)}$$

$$\therefore V_2 = 62.88V$$

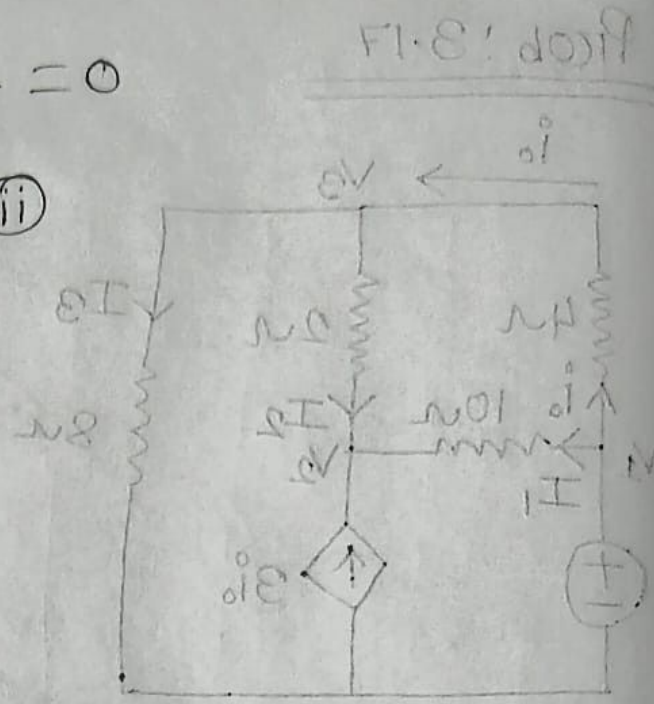
$$\therefore V_3 = 53.07V$$

$$i_o = \frac{60 - V_3}{4}$$

$$= \frac{60 - 53.07}{4}$$

$$\therefore i_o = 1.73A$$

Am



Apply KCL at Node \$V_3\$

$$0 = i_3 + i_2 + i_1$$

$$0 = \frac{60 - V_3}{4} + \frac{V_3 - 0}{2} + \frac{V_3 - 0}{2}$$

$$0 = 150 - 3V_3 + 10V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

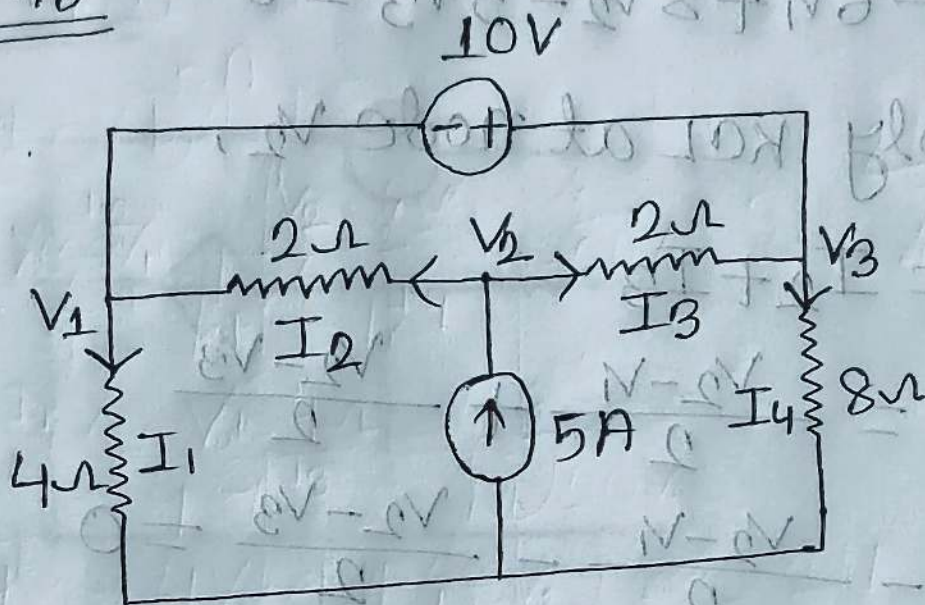
$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

$$0 = 150 - 3V_3 + 10V_3 - 10V_3 + 10V_3 - 10V_3$$

Рроб 3.18



$$V_3 - V_1 = 10$$

$$\Rightarrow -V_1 + 0 \cdot V_2 + V_3 = 10 \quad \text{--- (1)}$$

Apply KCL at super node,

$$I_2 + I_3 = I_1 + I_4$$

$$\Rightarrow \frac{V_2 - V_1}{2} + \frac{V_2 - V_3}{2} = \frac{V_1 - 0}{4} + \frac{V_3 - 0}{8}$$

$$\Rightarrow \frac{V_2 - V_1}{2} + \frac{V_2 - V_3}{2} - \frac{V_1}{4} - \frac{V_3}{8} = 0$$

$$\Rightarrow \frac{4V_2 - 4V_1 + 4V_2 - 4V_3 - 2V_1 - V_3}{8} = 0$$

$$\Rightarrow -6V_1 + 8V_2 - 5V_3 = 0 \quad \text{--- (ii)}$$

Apply KCL at node V_2 ,

$$5 = I_2 + I_3$$

$$\Rightarrow 5 = \frac{V_2 - V_1}{2} + \frac{V_2 - V_3}{2}$$

$$\Rightarrow 5 - \frac{V_2 - V_1}{2} - \frac{V_2 - V_3}{2} = 0$$

$$\Rightarrow \frac{10 - V_2 + V_1 - V_2 + V_3}{2} = 0$$

$$\Rightarrow V_1 - 2V_2 + V_3 = -10 \quad \text{--- (iii)}$$

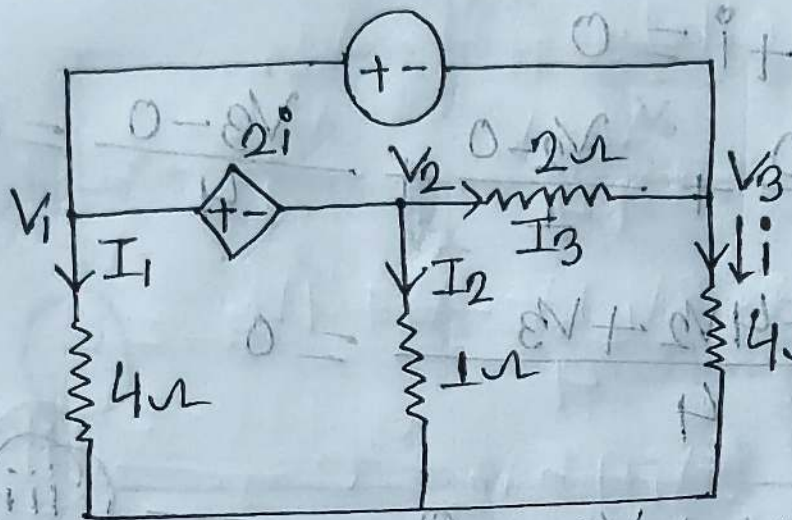
$$\therefore V_1 = 10V$$

$$\therefore V_2 = 20V$$

$$\therefore V_3 = 20V$$

Ans

Prob 3.20



$$i = \frac{V_3 - 0}{4}$$

$$2i = 2 \times \left(\frac{V_3 - 0}{4} \right)$$

$$= \frac{2V_3}{4}$$

$$V_1 - V_3 = 12$$

$$\Rightarrow V_1 + 0 \cdot V_2 - V_3 = 12 \quad \text{--- (i)}$$

$$V_1 - V_2 = 2i$$

$$\Rightarrow V_1 - V_2 = \frac{2V_3}{4}$$

$$\Rightarrow V_1 - V_2 - \frac{2V_3}{4} = 0 \quad \text{--- (ii)}$$

Apply KCL at super node, o.e.e don't

$$I_1 + I_2 + i = 0$$

$$\Rightarrow \frac{V_1 - 0}{4} + \frac{V_2 - 0}{4} + \frac{V_3 - 0}{4} = 0$$

$$\Rightarrow \frac{V_1 + 4V_2 + V_3}{4} = 0$$

$$\Rightarrow V_1 + 4V_2 + V_3 = 0$$

$$\therefore V_1 = -3V$$

$$\therefore V_2 = 4.5V$$

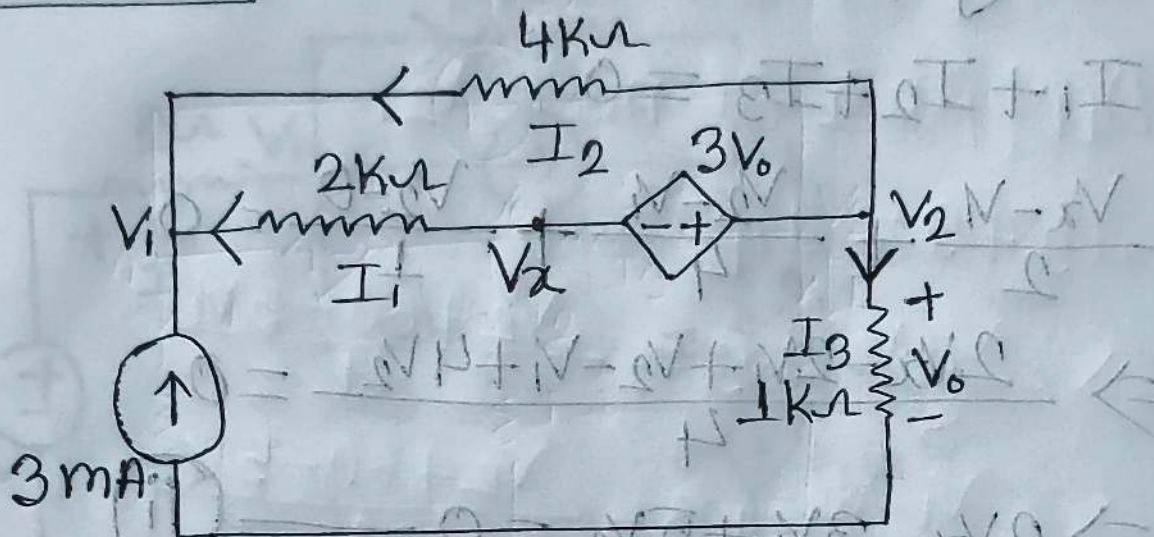
$$\therefore V_3 = -15V$$

Am

(iii)

$$0 = \frac{0-8V}{N} - 2V - N \leq$$

Prob 3.21



Determine V_1 and V_2

$$V_0 = 1 I_3$$

$$= 1 \times \frac{V_2 - 0}{1}$$

$$\therefore V_0 = V_2$$

$$3V_0 = 3V_2$$

$$-V_x + V_2 = 3V_2$$

$$\Rightarrow -V_x + V_2 - 3V_2 = 0$$

$$\Rightarrow -V_x - 2V_2 = 0$$

$$\Rightarrow -V_x + 0 \cdot V_1 - 2V_2 = 0$$

$$\begin{array}{l} V_x = -2V_2 \\ V_1 = 0 \\ V_0 = V_2 \end{array}$$

Apply KCL at super node, e.g. do it!

$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_2 - V_1}{2} + \frac{V_2 - V_1}{4} + \frac{V_2 - 0}{1} = 0$$

$$\Rightarrow \frac{2V_2 - 2V_1 + V_2 - V_1 + 4V_2}{4} = 0$$

$$\Rightarrow 2V_2 - 3V_1 + 5V_2 = 0$$

Apply KCL at node V_1

$$3 + I_1 + I_2 = 0$$

$$\Rightarrow 3 + \frac{V_2 - V_1}{2} + \frac{V_2 - V_1}{4} = 0$$

$$\Rightarrow \frac{12 + 2V_2 - 2V_1 + V_2 - V_1}{4} = 0$$

$$\Rightarrow 2V_2 - 3V_1 + V_2 = -12$$

$$\therefore V_2 = -6V$$

$$\therefore V_1 = 1V$$

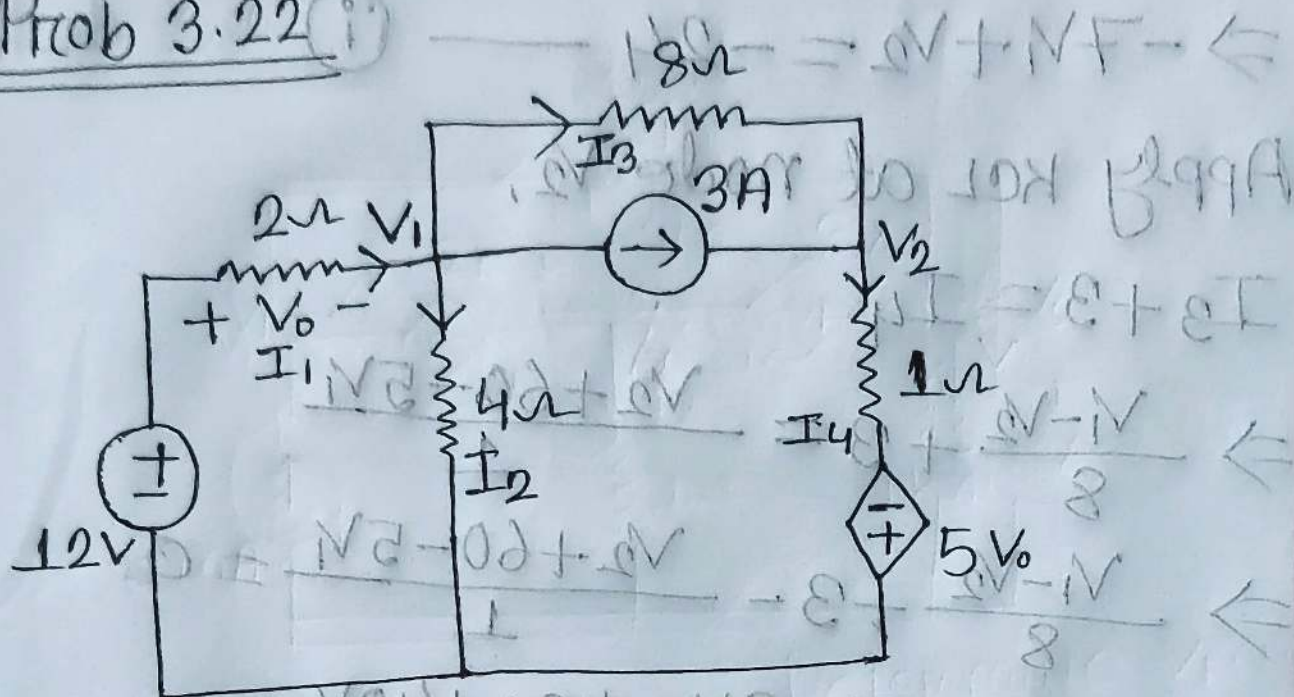
$$\therefore V_2 = 3V$$

$$V_0 = V_2$$

$$\therefore V_0 = 3V$$

Am

Prob 3.22 (i)



$$V_0 = 12 - V_1$$

$$5V_0 = 5(12 - V_1) \\ = 60 - 5V_1$$

Apply KCL at node V_1

$$I_1 + I_2 = I_3 + 3$$

$$\frac{12 - V_1}{2} + \frac{V_1 - 0}{4} = \frac{V_1 - V_2}{8} + 3$$

$$\Rightarrow \frac{12 - V_1}{2} + \frac{V_1}{4} - \frac{V_1 - V_2}{8} - 3 = 0$$

$$\Rightarrow \frac{48 - 4V_1 - 2V_1 - V_1 + V_2 - 24}{8} = 0$$

$$\Rightarrow -7V_1 + V_2 = -24$$

Prob 3.25

Apply KCL at node V_2 ,

$$I_3 + 3 = I_4$$

$$\Rightarrow \frac{V_1 - V_2}{8} + 3 = \frac{V_2 + 60 - 5V_1}{1}$$

$$\Rightarrow \frac{V_1 - V_2}{8} + 3 - \frac{V_2 + 60 - 5V_1}{1} = 0$$

$$\Rightarrow \frac{V_1 - V_2 + 24 - 8V_2 - 480 + 40V_1}{8} = 0$$

$$\Rightarrow 41V_1 - 9V_2 = 456$$

$$\therefore V_1 = -10.90V$$

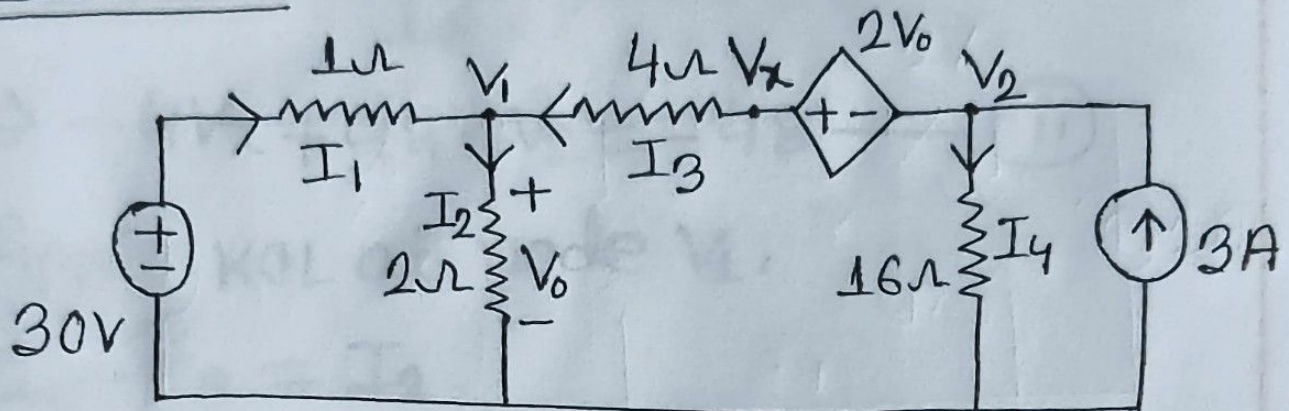
$$\therefore V_2 = -100.36V$$

Ans

$$0 = 8 - \frac{V_1 - V_2}{8} - \frac{V_2}{1} + \frac{V_2 + 60 - 5V_1}{1}$$

$$0 = \frac{48 - V_1 + V_2 - 8V_2 - 480 + 40V_1}{8}$$

Prob 3.23



Find $V_0 = ?$.

$$V_0 = V_1 - 0$$

$$\Rightarrow V_0 = V_1$$

$$V_x - V_2 = 2V_0$$

$$\Rightarrow V_x - V_2 = 2V_1$$

$$\Rightarrow V_x - 2V_1 - V_2 = 0 \quad \text{--- (i)}$$

Apply KCL at Super node,

$$3 = I_3 + I_4$$

$$\Rightarrow 3 = \frac{V_x - V_1}{4} + \frac{V_2 - 0}{16}$$

$$\Rightarrow 3 - \frac{V_x - V_1}{4} - \frac{V_2}{16} = 0$$

$$\Rightarrow \frac{48 - 4V_2 + 4V_1 - V_2}{16} = 0$$

$$\Rightarrow -4V_2 + 4V_1 - V_2 = -48 \text{ ——— (ii)}$$

Apply KCL at node V_1 ,

$$I_1 + I_3 = I_2$$

$$\Rightarrow \frac{30 - V_1}{1} + \frac{V_2 - V_1}{4} = \frac{V_1 - 0}{2}$$

$$\Rightarrow \frac{30 - V_1}{1} + \frac{V_2 - V_1}{4} - \frac{V_1 - 0}{2} = 0$$

$$\Rightarrow \frac{120 - 4V_1 + V_2 - V_1 - 2V_1}{4} = 0$$

$$\Rightarrow V_2 - 7V_1 = -120$$

$$\Rightarrow V_2 - 7V_1 + 0 \cdot V_2 = -120 \text{ ——— (iii)}$$

$$\therefore V_2 = 36.41 \text{ V}$$

$$\therefore V_1 = 22.34 \text{ V}$$

$$\therefore V_2 = -8.27 \text{ V}$$

$$V_o = V_1$$

$$\therefore V_o = 22.34 \text{ V.}$$

Am